Experiment 4: Implement half-ADDERS and Full-ADDERS

Theory:

A combinational logic circuit that performs the addition of two data bits, A and B, is called a half-adder. Addition will result in two output bits; one of which is the sum bit, S, and the other is the carry bit, C. The Boolean functions describing the half-adder are: S =A ⊕ B C = A B. A logic circuit for the addition of two one bit numbers is referred to as a half adder. A half adder can be constructed using EX-OR(IC-7486) and AND (IC7408) gates. The truth table shows, that in the first three rows there is no carry, whereas in the fourth row a carry is present. From the truth table, we obtain the logical expression for SUM and CARRY as, SUM=A'B+AB' CARRY=AB Half Adder/Subtractor is a basic ckt. that adds / subtracts 2 bits and generates sum or difference along with Carry / Borrow. Unlike half adder or subtractor a full adder / subtractor has the provision to take consideration of previous carry / borrow also.

Apparatus: Bread board , wires IC-7486(EX-OR),7408(AND)

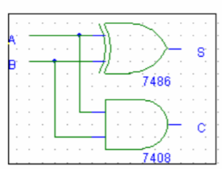
TRUTH TABLE

INPUTS		OUTPUTS	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

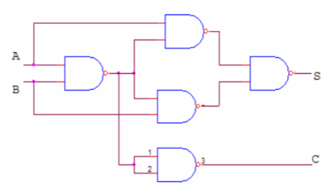
BOOLEAN EXPRESSIONS:

S=A ⊕ B
C=A B

i) Basic Gates

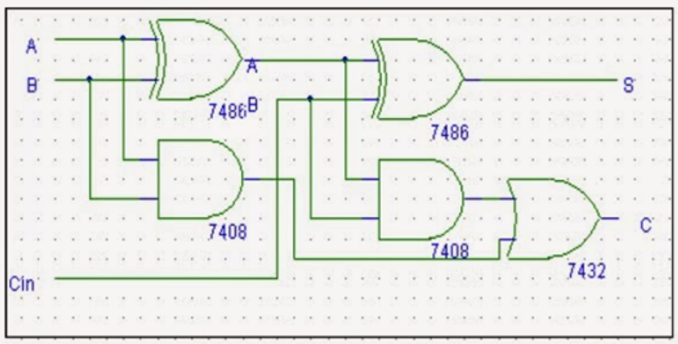


ii) NAND Gates



Full adder circuit diagram:

FULL ADDER CIRCUIT DIAGRAM :



FULL ADDER TRUTH TABLE

INPUTS			OUTPUTS	
A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Apparatus: Bread board, wires IC-7486(EX-OR),7408(AND), NOR(7402) and NAND (7400)